Kenneth W. Bingham

Senior Full-Stack Developer | Software Architect | AI-Directed Engineer

St. George, UT (Mountain Time) | Remote, or hybrid in St. George | Contact: butterflyfx.us/contact.html

LinkedIn: linkedin.com/in/kenneth-bingham-11378b10 | GitHub: github.com/kenbin64

Portfolio: butterflyfx.us | Games: playbutterflyfx.com | Gallery: kensartgallery.com | Videos: youtube.com/@kbingh | Substack: substack.com/@kbingh

SUMMARY

Senior full-stack developer, software architect and AI-directed engineer with 15+ years building enterprise, government, financial, logistics and avionics systems. Expert in Java, Spring Boot, C#, .NET Core, Angular, SQL, Azure and microservices. Directs AI as an engineering lead: specification first, every change reviewed, and tests that try to break each claim before anything ships. Strong foundation in discrete math, geometry, trigonometry, 3D graphics and spatial data, with hands-on work in RAG, embeddings, agents and LLM evaluation.

SKILLS

Languages: Java, C#, Python, JavaScript, TypeScript, SQL, HTML5, CSS3

Back end: Spring Boot, Spring Security, Spring Data JPA, Hibernate, JDBC, .NET Core, .NET 6, Razor Pages, Node.js, Express, FastAPI, REST APIs, microservices, MVC, WebSockets, JSP, Thymeleaf

Java persistence and REST: JPA and Hibernate entities (@Entity, @Table, @Id, @Column), relationship mapping (@ManyToMany, @OneToMany, @ManyToOne), lazy and eager fetching, Spring Data repositories, service layers, REST controllers (GET, POST, PUT, DELETE), HTTP request and response handling, Spring MVC

Design patterns: MVC, Singleton, Builder, Factory, Observer, publish-subscribe, repository and service layers, dependency injection

Front end and web design: Angular, AngularJS, React, GWT, jQuery, Bootstrap, responsive and mobile-first design, component architecture, semantic HTML, accessibility, performance and page-speed tuning, on-page and technical SEO, structured data

Databases: SQL Server, Oracle, PostgreSQL, pgvector, MySQL, MongoDB, MarkLogic (XQuery, XPath), SQLite; normalization, advanced joins, stored procedures, triggers, indexing, query optimization, schema design, data modeling, ETL

AI engineering: Claude Code in VS Code, agentic and multi-agent workflows, custom Model Context Protocol (MCP) server, MCP connected to an Ubuntu VPS, retrieval-augmented generation (RAG), embeddings and semantic search, prompt engineering, tool and function calling, guardrails, hallucination detection and mitigation, LLM evaluation (pass@k, blind pairwise ranking, rubric grading), adversarial benchmark design, CP-SAT constraint solving, token and cost optimization, private multi-model AI with Ollama and LM Studio

AI tools: Claude, ChatGPT, Gemini, Google AI Studio, GitHub Copilot, Cursor, Meshy, OpenArt

3D and spatial: Three.js, WebGL, WebXR, Unity, GameMaker, Blender, shaders, scene graphs, camera systems; voxel grids, octrees, quadtrees, spatial hashing, 3D coordinate indexing

Cloud and DevOps: Azure, Azure DevOps, AWS, Cloudflare Workers and KV, GitHub Actions, Docker and Docker Desktop, PowerShell, CI/CD, secrets management, Ubuntu Server installation and complete server management, Windows Server, Apache HTTP Server and nginx virtual hosts, Apache Tomcat, Let's Encrypt TLS, PM2, systemd, SSH hardening, Git push-to-deploy, versioned releases with rollback, Splunk, TIBCO BW

Build and package management: Maven and Gradle (Java), NuGet (.NET), npm (JavaScript), pip (Python), Webpack, multi-module Java projects; Git, GitHub, GitLab, Bitbucket, TFS, SVN

Testing: JUnit and TestNG (Java), MSTest, NUnit and xUnit (.NET), Selenium WebDriver and Grid parallel testing, Postman, Puppeteer headless-browser testing, automated regression suites, deterministic test harnesses

API, debugging and diagnostics: Postman, Swagger and OpenAPI (Swashbuckle in ASP.NET Core), JSON request and response inspection, browser developer tools (Chrome DevTools: network, console, DOM, performance), Visual Studio, IntelliJ and VS Code debuggers (breakpoints, watches, stepping), log analysis with Splunk and application and server logs

Design and data: Photoshop, Visio, UML, architecture and ERD diagrams, Excel and Google Sheets (pivot tables, XLOOKUP, INDEX/MATCH)

IDEs: IntelliJ IDEA, WebStorm, Android Studio, Eclipse, Visual Studio, VS Code

EXPERIENCE

AI Engineer, Evaluation and Reliability (Contract)

Handshake AI, Remote | Jul 2026 to Present

- Author adversarial benchmark tasks for Terminal-Bench 2 that defeat frontier coding agents, verified at pass@5 0/5: the reference agent at maximum reasoning effort fails every attempt.
- Built an offline difficulty-certification harness (candidate generator, CP-SAT constraint-solver filter and a blind cold-agent pass@2 run) that certifies task difficulty in minutes instead of a 68-minute CI cycle.
- Wrote a deterministic pre-submit linter encoding 104 authoring rules that catches solution leaks, ambiguous framing and self-verifiable traps before a task reaches review.
- Designed two pre-build gates, non-self-verifiable and non-internet-lookupable, that guard against metric gaming and training-set contamination.
- Run blind pairwise LLM evaluations (anonymized dual-model trajectories, forced-choice rating, grading on correctness, behavior, communication and code style), and evaluate geometric reasoning and visual question answering.
- Root-caused an MSYS path-conversion defect that silently misplaced trajectory exports inside the Docker VM, replacing the broken scripts with PowerShell tooling; reported platform-wide CI defects, including a pull_request_target fork-checkout failure.

Software Developer

DTSI, Hill Air Force Base, UT | Apr 2024 to May 2026

- Principal developer on Air Force logistics and operational systems in Java, Spring Boot, Hibernate, Angular and Bootstrap, and in C#, .NET Core Razor Pages and SQL Server, under strict DoD change control.
- Configured CI/CD pipelines and secure deployment workflows with TFS version control, with zero rollbacks.
- Built enterprise dashboards and automated operational workflows that streamlined internal processes and reduced technical debt.
- Redesigned user interfaces with graphical buttons, icons and modern visual design, driving higher adoption and daily use.

Software Developer

Humana, Remote | Feb 2022 to Jan 2024

- Built .NET 6 services in C# and Visual Studio, and HL7 and FHIR interoperability pipelines for high-volume insurance claims under HIPAA.
- Modernized legacy applications from .NET Framework to .NET Core (.NET 6) for performance and cross-platform support.
- Designed doctor-to-patient-to-insurer communication pipelines, and documented the architecture in Visio.
- Built Python ETL automation and Linux workflows that removed manual processing, and optimized SQL for large-scale claims data.

Software Developer

Software Implementers, Hill Air Force Base, UT | Apr 2021 to Dec 2021

- Developed Java, Spring, Spring Boot and Hibernate services and responsive Angular and Bootstrap front ends for aircraft-parts testing, maintenance and asset tracking.
- Turned the testers' manual spreadsheets into a central web-based CMS and workflow platform for parts tracking, test protocols and maintenance.
- Integrated hardware scanning equipment with the software for inventory, data entry and asset tracking.
- Built dashboards and workflow automation, and designed architecture diagrams, database schemas and data flows in Visio. Held a U.S. Air Force Secret clearance.

Software Developer (Contract via ConsultNet)

Client: The Church of Jesus Christ of Latter-day Saints, Salt Lake City, UT | Sep 2020 to Oct 2020

- Contributed to an internal application for tracking and versioning Church leaders' talks, in Java, Spring Boot, Hibernate, JSP, Thymeleaf and Bootstrap.
- Implemented normalized schemas with advanced SQL join logic and CRUD features; configured secure authentication and role-based access.

Software Developer

Gwinnett County Public Schools, Lawrenceville, GA | Feb 2020 to Aug 2020

- Built automated test frameworks in Java and Selenium WebDriver for the district's cloud platforms, with Azure DevOps pipelines.
- Built district administrative and reporting tools with secure workflows and Bootstrap dashboards.

Java Developer (Contract via Collabera)

Norfolk Southern Railway, Atlanta, GA | Apr 2019 to Aug 2019

- Built TIBCO BusinessWorks and Spring Boot messaging for locomotive fuel-efficiency optimization.
- Used Splunk for production diagnostics and Jackson for XML handling.

Java Developer, Test Automation (Contract via Collabera)

IBM, Suwanee, GA | Nov 2018 to Mar 2019

- Worked with the architect and program manager to build a Selenium WebDriver framework that uses parent-child relationships to find groups of elements from a single element, significantly reducing boilerplate and search time.
- Built grid-based parallel testing and produced test reports in Azure.

Java Developer

InComm, Taylorsville, UT | Apr 2018 to Aug 2018

- Developer in test for payment and prepaid card software in Java, JavaScript and Angular 6, automating with Selenium WebDriver, TestNG and Postman.
- Planned sprints, contributed to software design, and streamlined processes to cut duplication and boilerplate; used AWS for image storage.

Java Developer (via Software Technology Group)

Finicity, a Mastercard company, Salt Lake City, UT | Sep 2017 to Apr 2018

- Built Spring Boot APIs for Finicity clients and Fannie Mae that retrieve credit information from the major credit reporting agencies, with pay and payment data, to produce reports for lenders.
- Created JSON endpoints for the front end, an OCR text-comparison application and a regular-expression validation application, on MySQL and MongoDB.
- Test-driven development in Java 8 with JUnit, Spring Boot, Spring Data, Hibernate and Spring Security; a white-labeled email system with SendGrid; reports with Apache PDFBox.

Java Developer

EFS, a WEX company, Ogden, UT | Feb 2017 to Sep 2017

- Developed and administered the customer portal for fleet fuel-card purchases in the trucking industry, in Java 8 Servlets and JSP with JavaScript, jQuery and Bootstrap, on Oracle and MySQL.

Java Developer

CUProdigy, Layton, UT | Apr 2015 to Feb 2017

- Developed the credit union core system for small and mid-size credit unions: teller applications, management and operations, record keeping, account management, loan processing, interest calculation, and payment and collection management.
- Built in Java, Spring, Spring Boot, Spring Data JPA, Hibernate, Tomcat and MySQL, with a GWT DataGrid and CellTable front end.
- Delivered the payment system and payment buckets, emailed PDF documents, document transfer over HTTP, account and loan applications, admin and configuration pages, and JasperReports reporting.

Java Developer, Adobe AEM 6.0 (Contract via ConsultNet)

Axis41, Salt Lake City, UT | Nov 2014 to Apr 2015

- Built the O.C. Tanner CMS in Adobe AEM 6.0, including a vanity-URL redirect tool in Java, Sling and Jackrabbit, and components and widgets in JavaScript and jQuery.
- Built Stanford Health Care site components in AEM 6.0 with Bootstrap 3 and jQuery UI, plus JSP and Java Servlet components, scroll animation, sprites and animated sliders.

Java Developer (Contract via TEKsystems)

USANA Health Sciences, West Valley City, UT | Jun 2014 to Oct 2014

- Front-end work in AngularJS, JavaScript, jQuery, HTML5, CSS3 and Bootstrap: a template that lets users build their own websites.
- Created AngularJS controllers and directives for widget data, sprite-based scroll animation, and animated sliders and transitions with jQuery UI and Backbone.

Software Consultant

Software Technology Group, Salt Lake City, UT | Sep 2011 to Jun 2014

- Consultant for STG clients in Java, Spring, JavaScript, jQuery, Bootstrap, AJAX, XML, XQuery, XPath, Swing, MarkLogic, Maven, Ivy, Ant, SQL and Flex.
- Missionary Portal for The Church of Jesus Christ of Latter-day Saints (Mar 2012 to Dec 2013): a web application used by thousands of missionaries worldwide for weekly training, reporting and information. Helped plan its architecture and multi-server deployment and built most of the UI: responsive Bootstrap 2 and 3 layouts, the weather app, a contact page with real-time validation, announcements, menus and a printable acceptance-letter page, with Backbone, Handlebars, RequireJS and Less, Java Spring and Hibernate web services to Oracle, and AngularJS for the Leadership Portal.
- Translation application for the same client (Sep 2011 to Mar 2013): Java Swing components for sending documents for translation into many languages, including custom tables, buttons and 2D-graphics panels, drag and drop between a tree and a table, custom sort and filter, and undo; also Flex 4 and ActionScript, with Maven, TFS and SQL Server.
- CourtView, for another STG client (Dec 2013 to Jun 2014): a web application used by courts around the country to track and schedule cases, fine payments and sentences. Built UI panels and tables with DataTables, jQuery, Bootstrap 3 and JSTL, and back-end work in Java, Spring and Hibernate, with Ivy, Ant and Git.
- Configured MarkLogic Server (databases, forests, HTTP, WebDAV and XDBC app servers) and wrote XQuery, XPath, XSLT transformations and DTD and XSD schemas.

Java Developer (Internship while completing my degree)

Boresight, Ogden, UT | Jun 2010 to Sep 2011

- Built a desktop application, installed in military and police aircraft, for navigation, sensor and camera integration in Java and Java Swing: interpreted binary joystick-control data for aircraft-mounted cameras and integrated real-time mapping with aviation waypoints.
- Created mobile mapping apps.

Customer Service Agent (College Role)

Juniper Networks | 2008 to 2009

- Troubleshot Juniper routers and enterprise networking: TCP/IP, routing tables, NAT, DHCP, VLANs and firewall rules.

SELECTED PROJECTS (HUMAN AND AI COLLABORATION)

In each project I set the vision, design, standards and priorities and verified every result; an AI coding assistant wrote and tested the code under my direction.

- **Agentic full-stack development pipeline.** VS Code and Claude Code connected through a custom MCP server (JSON-RPC 2.0); the agent builds full-stack applications and runs deployments under my direction, with versioned releases and one-command rollback.
- **Chomp 3D: Wally's Cairns Adventure, playbutterflyfx.com/chomp.** Seven-level 3D browser game for desktop, phone and VR (WebXR). 1,022 unit checks and 107 browser tests gate 14 versioned releases with one-command Cloudflare rollback; Workers leaderboard and feedback form.
- **FastTrack 3D, playbutterflyfx.com/fasttrack.** 3D card-driven board game with bots at three difficulties. Fixed a multiplayer turn desync with host-authoritative turn control over WebSockets, proven by a headless two-client regression test.
- **citeline, github.com/kenbin64/citeline.** Retrieval-augmented generation over federal regulations with PostgreSQL, pgvector embeddings and FastAPI: answers with a citation or abstains. 31 unit tests and a 28-case evaluation set; a live version runs on Cloudflare Workers.
- **mcp-gateway, github.com/kenbin64/mcp-gateway.** Audited MCP server in Python and FastAPI: a closed list of read-only tools, schema-validated arguments, every call logged including refusals, and errors that cannot leak. 29 tests with ruff and mypy.
- **Ken's Art Gallery, kensartgallery.com.** Walk-through 3D gallery of 59 works with rooms that lay themselves out, a Python image pipeline (watermark, signature, duplicate detection), code-generated audio and a Cloudflare Workers contact form.
- **Cost of Saving, costofsaving.com.** Decision calculators on Cloudflare Workers with every formula published: 517 automated checks across 13 suites.
- **Demystifying AI (video series), youtube.com/@kbingh.** Standalone videos explaining AI from "What is AI?" to advanced topics, including how to spot and reduce hallucinations.
- **Chomp! (original Android game).** Conceived, designed, programmed and animated the original Chomp!, published on Google Play, in Java with the LibGDX game engine, Photoshop and Illustrator; all artwork my own. Chomp 3D is its sequel.
- **ButterflyFX geometry engine and 3D grid systems.** Research on hyperbolic-paraboloid (z = xy) geometry engines, voxel storage and 3D spatial indexing: store the rule and derive the data on demand. Also a Connect Four variant on a Schwarz Diamond lattice.

EDUCATION

B.S., Computer Science and Software Engineering, Weber State University, Ogden, UT, 2011

SECURITY CLEARANCE

U.S. Air Force Secret clearance (previously held), with years of delivery inside DoD environments at Hill Air Force Base.

INTERESTS

Drumming, painting, singing, 3D modeling and 3D printing, game development, AI-generated art and video.